

Cloud LearnX

Mathematics

Circle Theorems — Practice Worksheet

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Topic: Circle Theorems

Instructions:

- Answer ALL questions.
- Show ALL working — method marks are awarded for correct steps.
- Write answers in the spaces provided.

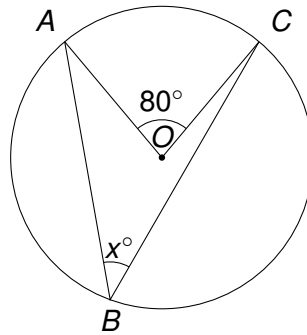
Answer ALL ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

1 A , B and C are points on the circumference of a circle with centre O .

The angle at the centre, AOC , is 80° . The angle at the circumference, ABC , is x° .

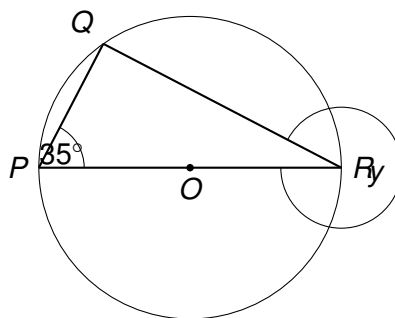


Find the size of angle x , giving a reason for your answer.

$x = \dots\dots\dots^\circ$
(2)

(Total for Question 1 is 2 marks)

- 2 The diagram shows the cross-section of a satellite dish, modelled as a circle with centre O . PR is a diameter of the circle, and Q is a point on the circumference. PQ and QR are chords forming a triangle inscribed in the circle.



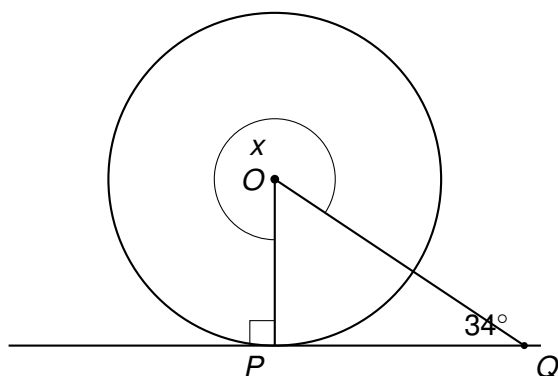
Angle $QPR = 35^\circ$.

Work out the size of angle y .

$y = \dots\dots\dots^\circ$
(2)

(Total for Question 2 is 2 marks)

- 3** A space telescope has a circular mirror, centre O . A support strut is bolted to the edge of the mirror at point P , and lies along the tangent to the circle at P . A second strut OQ connects the centre O to a fixing bracket at Q on the tangent line, as shown.

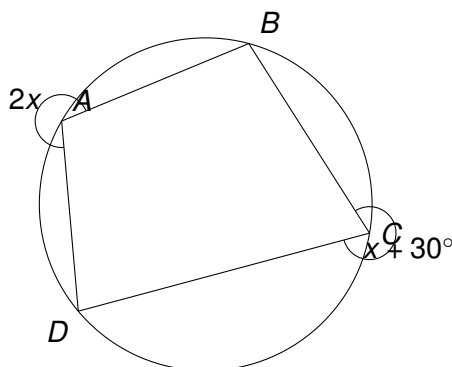


The angle between the strut OQ and the tangent line at Q is 34° .
Calculate the size of angle x , marked at the centre O .

$x = \dots\dots\dots^\circ$
(3)

(Total for Question 3 is 3 marks)

- 4 A , B , C and D are four points on the circumference of a circle, forming the cyclic quadrilateral $ABCD$. Angle $DAB = 2x$ and angle $BCD = x + 30^\circ$, as shown in the diagram.

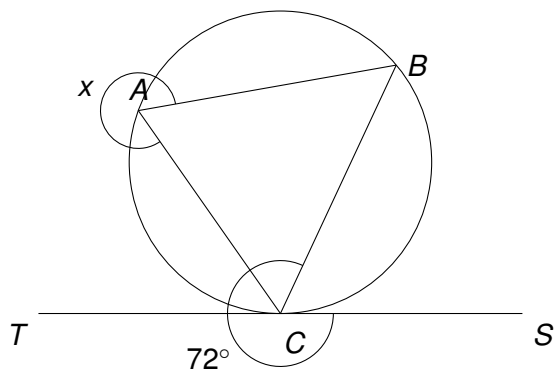


Find the value of x and hence work out the size of each of the angles DAB and BCD .

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(3)

(Total for Question 4 is 3 marks)

5 A , B and C are points on the circle. TCS is a tangent to the circle at C .



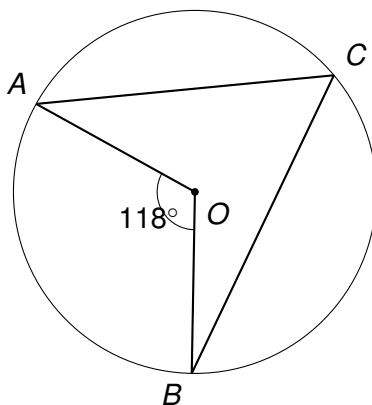
The angle between the tangent TCS and the chord CB is 72° , as shown.
Find the size of angle x , giving a reason for your answer.

$x = \dots\dots\dots^\circ$
(3)

(Total for Question 5 is 3 marks)

- 6 A space telescope has a circular lens with centre O . Points A and B lie on the circumference of the lens, and OA and OB are radii marking the edges of a support bracket. Point C also lies on the circumference, on the major arc AB , and represents the edge of the telescope's field of view.

Angle $AOB = 118^\circ$.



- (a) Find the size of angle OAB .

$$OAB = \dots\dots\dots^\circ$$

(2)

- (b) Work out the size of angle ACB , the angle representing the telescope's field of view at C .

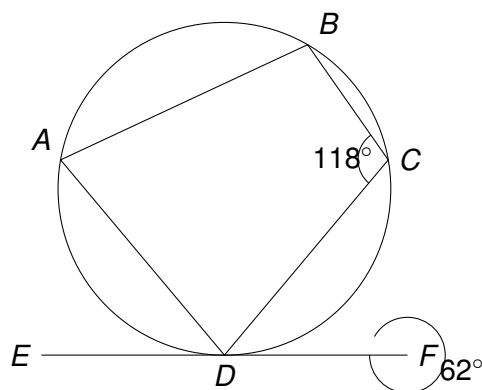
$$ACB = \dots\dots\dots^\circ$$

(2)

(Total for Question 6 is 4 marks)

7 A , B , C and D are four points on the circumference of a circle. The line EDF is a tangent to the circle at point D , where E and F lie on either side of D on the tangent line.

$ABCD$ is a cyclic quadrilateral. Angle $BCD = 118^\circ$ and angle $ADF = 62^\circ$.



Show that angle $ADC = 118^\circ$.

You must show each step of your working and state the circle theorems you use.

(4)

(Total for Question 7 is 4 marks)

TOTAL FOR PAPER IS ?? MARKS